

# Laplacian-Eigenmode Static Sources to Gluonic Flux-Tube Profiles

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# Project aim: from static potential to gluonic flux-tube profiles

## Physics target

- ▶ Build static  $Q\bar{Q}$  sources with Laplacian-eigenmode trial states.
- ▶ Validate via a static-potential benchmark.
- ▶ Then insert local gluonic probes to map the flux tube.

## Gluonic probe: action density

$$S(\vec{z}, t) = \frac{1}{2} \left( E^2(\vec{z}, t) + B^2(\vec{z}, t) \right),$$

$$\Delta S_{Q\bar{Q}}(\vec{z}; R, \tau) = \frac{\langle L(R, \tau) S(\vec{z}, t_{\text{mid}}) \rangle}{\langle L(R, \tau) \rangle} - \langle S(\vec{z}, t_{\text{mid}}) \rangle.$$

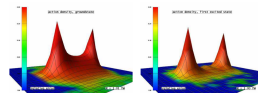


Figure 3: Action density distribution for the ground state and the first excitation.

### 4. Conclusion

At any distance the action density looks like a superposition between (hypothetical)  $Q\bar{Q}$  and  $q\bar{q}$  distributions with little interference: light pair creation seems to occur non-localized and instantaneously. Applying the decay model of Ref. [5], where the interaction term is instantaneous and only depends on the separation  $r$ , we undershoot the experimental  $T(4S) \rightarrow B\bar{B}$  decay rate by a factor of about two. This appears very reasonable, considering the crudeness of the model and the fact that the gap  $\Delta E_c$  will increase with lighter, more realistic sea quark masses. We are studying the situation in more detail.

### Acknowledgments

Flux-tube/string-breaking paper: Bali et al., PoS LAT2005 (2005).

Given a static quark and antiquark separated by  $R$ , how is the gluonic field modified at the point  $\vec{z}$ ?

# Methodology and resources used in the project

## Operator construction

- ▶ Distillation: low-rank smooth subspace from 3D covariant Laplacian eigenvectors.
- ▶ Laplace static trial states: replace spatial Wilson lines by eigenvector outer products.
- ▶ Static perambulators: precompute time propagation between Laplacian modes.

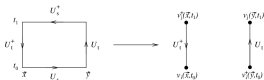


FIG. 1: The spatial Wilson lines  $U_s(\vec{x}, \vec{y}, t)$  of the classical Wilson loop  $W(R, T)$  of size  $(R = |\vec{y} - \vec{x}|) \times (T = |t_1 - t_0|)$  (left) can be replaced by Laplacian eigenvector pairs  $v_l(\vec{x}, t)v_l^*(\vec{y}, t)$  (right, eigenvector pairs to be read in anti-clockwise direction), to form Laplace trial state correlators via the two static perambulators  $\tau_{ij}(\vec{x}, t_0, t_1)$  and  $\tau_{ij}(\vec{y}, t_0, t_1)$ .

III. THE STATIC QUARK-ANTI-QUARK POTENTIAL over the mass-plateau region of the fractional overlap

## Analysis and statistics

- ▶ Validate  $L(R, \tau)$  via  $V_{\text{eff}}(R, \tau)$  and  $V(R)$  before local-probe ratios.
- ▶ Action/energy density from  $E^2$  and  $B^2$  insertions.
- ▶ pyerrors/Gamma-method error analysis; jackknife cross-checks.

|         |                                                           |
|---------|-----------------------------------------------------------|
| 2009    | Peardon et al.: distillation / Laplacian modes            |
| 2023    | Höllwieser et al.: static $Q\bar{Q}$ Laplace trial states |
| 2005    | Bali et al.: string breaking and action-density profiles  |
| 2004–06 | Wolff: Gamma method for Monte Carlo errors                |

# Current progress: from Laplace-static correlator to potential

## Laplacian eigenmodes

$$\Delta v_i = \lambda_i v_i, \quad i = 1, \dots, N_v.$$

## Static perambulator

$$\tau_{ij}(\vec{x}; t_0, t_1) = v_i^\dagger(\vec{x}, t_0) U_t(\vec{x}; t_0, t_1) v_j(\vec{x}, t_1).$$

## Laplace-static correlator

$$L(R, \tau) = \left\langle \sum_{i,j=1}^{N_v} \rho_i^2 \rho_j^2 \tau_{ij}(\vec{y}; t_0, t_1) \tau_{ji}(\vec{x}; t_1, t_0) \right\rangle,$$

$$R = |\vec{y} - \vec{x}|, \quad \tau = t_1 - t_0.$$

## Effective energy

$$V_{\text{eff}}(R, T) = \log \frac{L(R, T)}{L(R, T+1)}.$$

## Current profile choice

$$\rho_i = 1, \quad i = 1, \dots, N_v, \quad N_v = 10.$$

## Cornell benchmark fit

$$aV(R) = aV_0 + \sigma a^2 R - \frac{\alpha}{R}.$$

Current tests:  $N_v = 10$  with constant eigenmode weights  $\rho_i = 1$ ; first validation through static-potential plateaus and Cornell fits.

## Implemented code path

- ▶ Read gauge fields and Laplacian eigenvectors from the existing QCD-library and HDF5 setup.
- ▶ Build  $\tau_{ij}(\vec{x}, t_0, t_1)$  for each chosen  $\tau = t_1 - t_0$ .
- ▶ Contract two static perambulators to form  $L(R, \tau)$ .
- ▶ Analyze  $V_{\text{eff}}$ , plateau windows, and correlated fits.

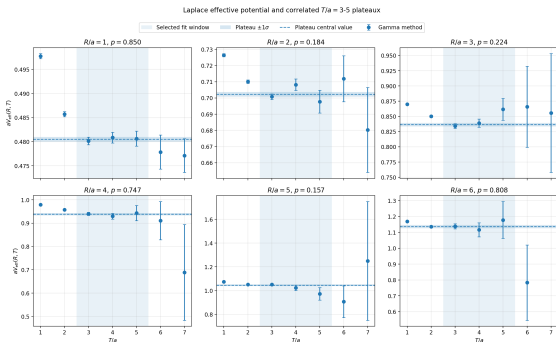
Main compute test:

`src/test_laplace_RT_scan.c`.

## Current production setup

- ▶  $N_{\text{cfg}} = 100$  gauge configurations.
- ▶  $N_v = 10$  eigenvectors with constant weights  $\rho_i = 1$ .
- ▶ Temporal-source average:  $t_0 = 0, 8, 16, 24, 32, 40$ .
- ▶ Validated scan:  
 $R/a = 1 \dots 6,$   
 $\tau/a = 1 \dots 8.$
- ▶ Diagnostic scan:  
 $R/a = 1 \dots 12,$   
 $\tau/a = 1 \dots 10.$

# Validation I: effective energies and plateau fits



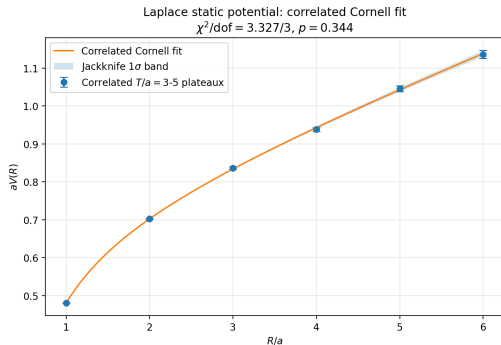
## Outcome

- Validated region:  $R/a = 1 \dots 6$ .
- Stable plateau window:  $\tau/a = 3 \dots 5$ .
- Correlated constant fits give acceptable  $\chi^2/\text{dof}$  and  $p$ -values.
- Validated  $L(R, \tau)$  normalizes the probe ratio.

**Takeaway:** source construction validated.  
Validated  $L(R, \tau)$  normalizes the local-probe ratio:

$$\Delta S_{Q\bar{Q}} = \frac{\langle LS \rangle}{\langle L \rangle} - \langle S \rangle.$$

# Validation II: static potential and Cornell fit



## Correlated Cornell fit

$$aV(R) = aV_0 + \sigma a^2 R - \frac{\alpha}{R},$$

$\chi^2/\text{dof} = 1.11, \quad p = 0.344.$

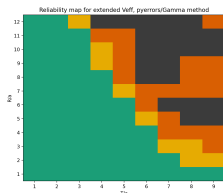
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|                  |             |
|------------------|-------------|
| $aV_0$           | 0.6646(82)  |
| $\sigma a^2$     | 0.08646(24) |
| $\alpha$         | 0.2705(60)  |
| $a\sqrt{\sigma}$ | 0.2940(40)  |

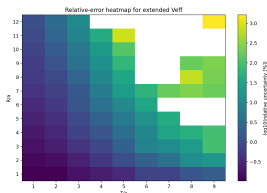
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- Potential rises as expected.
- Fit stable under moderate range checks.
- Confirms expected static-potential behavior.

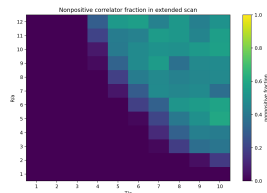
# Reliability diagnostics for the extended scan



**Reliability map**  
combined quality class



**Relative uncertainty**  
statistical noise level



**Non-positive fraction**  
sign/log stability check

## Good

low uncertainty;  
positive samples

## Marginal

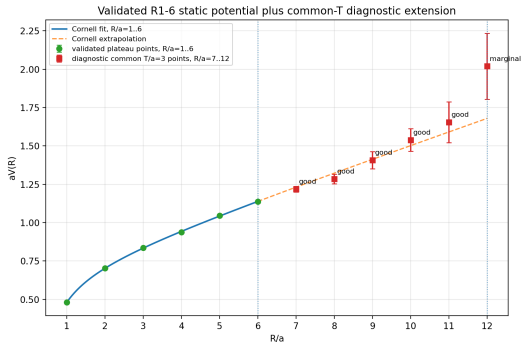
usable mainly for  
trend checks

## Diagnostic/invalid

large uncertainty or  
sign problems

The extended scan is used to select safe parameter windows for the first flux-tube probe measurements.

# Extended-distance diagnostic scan



Red points indicate a common  $\tau/a = 3$  diagnostic estimate beyond the validated fit range.

## What this tells us

- ▶ Extended scan reproduces the validated  $R/a = 1 \dots 6$  region.
- ▶  $R/a = 7, 8$  show selected plateau-like windows.
- ▶ Larger  $R/a$  values remain diagnostic with the current statistics/profile.

# Next step: local gluonic density insertion

## Field-strength probe

$F_{\mu\nu}(x) \leftarrow$  clover/plaquette estimator

from oriented  $\mu\nu$  loops.

$$E^2(\vec{z}, t) = \sum_{i=1}^3 \text{Tr} [F_{it}(\vec{z}, t)^2],$$

$$B^2(\vec{z}, t) = \sum_{i < j} \text{Tr} [F_{ij}(\vec{z}, t)^2],$$

$$S(\vec{z}, t) = \frac{1}{2} (E^2(\vec{z}, t) + B^2(\vec{z}, t)).$$

## Physical question

How is the local gluonic density modified by a static  $Q\bar{Q}$  pair separated by  $R$ ?

## Normalized and vacuum-subtracted profile

$$C_L(R, \tau) \equiv \langle L(R, \tau) \rangle,$$

$$C_{LS}(\vec{z}; R, \tau) \equiv \langle L(R, \tau) S(\vec{z}, t_{\text{mid}}) \rangle.$$

$$\Delta S(\vec{z}; R, \tau) = \frac{C_{LS}(\vec{z}; R, \tau)}{C_L(R, \tau)} - \langle S(\vec{z}, t_{\text{mid}}) \rangle.$$

## Prototype choices

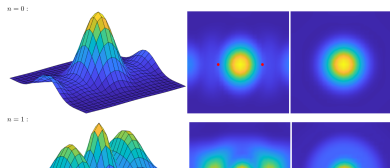
- ▶ insert probe at  $t_{\text{mid}} = t_0 + \tau/2$  where possible
- ▶ first source pair: on-axis  $R/a = 3$  or 4
- ▶ start with  $S$ , then analyze  $E^2$  and  $B^2$  separately

# Near-term plan for flux-tube profiles

## Immediate prototype

1. Reuse the validated  $L(R, \tau)$  pipeline and temporal source averaging.
2. Insert local  $S(\vec{z}, t_{\text{mid}})$  using qcdlib field-strength routines.
3. Accumulate  $\langle LS \rangle$ ,  $\langle L \rangle$ , and  $\langle S \rangle$  in one pass.
4. First output a 2D mid-plane map; then generalize to full 3D HDF5 data.

First target: on-axis  $R/a = 3$  or  $4$  in the validated plateau window.



Analogy from Laplace trial-state profiles:  
later we aim to map local gluonic density  
instead.

First physics map:  $\Delta S(\vec{z}; R, \tau)$ .  
Later:  $\Delta E^2$ ,  $\Delta B^2$ , smeared probes,  
and orientation averaging.

# Status summary: validated source and next probes

## Validated baseline

- ▶ Static perambulator and Laplace-static correlator pipeline is working.
- ▶ Six temporal-source averaging is implemented.
- ▶  $R/a = 1 \dots 6$  potential extraction is validated with correlated plateau fits.
- ▶ Cornell benchmark gives a physically sensible and statistically acceptable potential.

## Next implementation step

- ▶  $R/a = 7 \dots 12$  scan is useful diagnostically, but not yet final physics.
- ▶ Flux-tube profiles require inserting a local field-strength probe.
- ▶ First target:  $\Delta S(\vec{z}; R, \tau)$  for on-axis  $R/a = 3$  or  $4$ .
- ▶ Open choices: probe smearing, orientation averaging, and statistics.

Current stage: static-source validation is complete; implementation now moves to local gluonic-density probes.

# Backup: selected files and commands to show

| Purpose              | File / result                                        |
|----------------------|------------------------------------------------------|
| Code                 | src/test_laplace_RT_scan.c                           |
| Plateaux             | laplace_Veff_correlated_plateau_T3-5_t0avg6_R1-6.png |
| Cornell fit          | laplace_static_potential_cornell_R1-6_t0avg6.png     |
| Reliability map      | laplace_reliability_map_t0avg6_R12T10.png            |
| Diagnostic extension | laplace_static_potential_R1-6_fit_plus_R7-12...png   |
| R7/R8 window check   | laplace_plateau_window_comparison_R7_R8...png        |

## Results and plots location on Stromboli

```
cd /home/m2130292/Masterarbeit/laplace_static
ls -lh results/figures/*t0avg6*.png | head
```